

**BS 7671:2018 + AMENDMENT 4:2026 · TABLES 41.2 – 41.5**

Earth Fault Loop Impedance — Z_e & Z_s

Every maximum Z_s in the Regulations on three sheets you can pin up — circuit-breakers, fuses, RCDs, and the withdrawn BS 3871 types still sitting in older boards

VALUES TABULATED

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ABOUT THIS REFERENCE

BASIS	$C_{min} = 0.95$, exactly as tabulated BS 7671 Reg 253.0 and Reg 543.4. Nothing here is pre-corrected or rescaled.
DEVICES COVERED	Circuit-breakers, fuses and RCDs BS EN 60898 · BS EN 61009 · BS 88-2 · BS 88-3 · BS 3036 · BS 1362 · BS 3871 Types 1–4
COLD MEASUREMENTS	Compare against $0.8 \times$ the tabulated figure The Appendix 3 acceptance criterion. Sheet 1 works an example through.
USE	Free to print, share and pin up On site, in the van or in the office.
ISSUED	2026-08-27
FULL REFERENCE	elec-mate.com/guides/maximum-zs-values-bs-7671 With a free Z_s lookup calculator.

Earth Fault Loop Impedance — Ze & Zs

Maximum permitted values · BS 7671:2018+A4:2026 · values as tabulated, C_{min} 0.95



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1 · External earth fault loop impedance (Ze)

Measured at the origin with the installation isolated and the main earthing conductor disconnected. For TN these are the distributor's typical maximums, and what the BS 7671 tables assume. **TT is different** — the return is through your own electrode, so 21 Ω is a working ceiling you confirm by measurement.

TN-S

0.80 Ω

Cable sheath / armour return

TN-C-S (PME)

0.35 Ω

Combined PEN conductor

TT

21 Ω

Earth electrode — RCD required

Not a statutory cap. Typical maximums for supplies up to 100 A. Ze must be measured and recorded at every installation — if your reading exceeds it, query the DNO.

2 · How Ze and Zs relate

$$Z_s = Z_e + (R_1 + R_2)$$

Ze is what the supply brings to your origin; (R₁+R₂) is what your circuit adds. Zs is the total the device sees under an earth fault — and it is Zs that must not exceed the tabulated maximum.

3 · Which disconnection time applies

Circuit	Time	Regulation
Final circuit ≤ 63 A with one or more socket-outlets	0.4 s	411.3.2.2
Final circuit ≤ 32 A supplying only fixed equipment	0.4 s	411.3.2.2
Distribution circuit, or final circuit above those ratings	5 s	411.3.2.3

4 · Reading the tables on sheets 2 and 3

Every figure on sheets 2 and 3 is the BS 7671 tabulated maximum, derived with C_{min} = 0.95 — as printed in the Regulations, nothing pre-corrected. C_{min} is the minimum voltage factor for variation in time and place and for transformer tap changes; for an LV supply under the ESQCR it takes the value 0.95 (Reg 253.0, Reg 543.4). So a Type B 32 A is 230 × 0.95 ÷ (5 × 32) = **1.37 Ω**. At C_{min} = 1.0 it would read 1.44 Ω — if your reference shows that, it is not the BS 7671 figure.

Temperature — and this is where most people trip up.

The tabulated values assume conductors at their operating temperature (70 °C for thermoplastic; NOTE 2 to Tables 41.2–41.4). You measure cold, so BS 7671 sets a **separate acceptance criterion in Appendix 3**. Regulation 411.4.4 (or 411.5.4) is met when:

$$Z_{s(\text{measured})} < 0.8 \times U_0 \div (I_a \times C_{\text{min}}) \text{ — in practice, } 0.8 \times \text{the tabulated figure}$$

GN3 1.16.9 restates it, which is why it gets called "the GN3 factor" — but it is BS 7671's own criterion, not merely guidance.

⚠ The 0.8 is NOT applied anywhere in this document. Apply it yourself whenever you compare a cold measurement.

5 · Worked example

Step	Working	Result
32 A Type B RCBO on TN-C-S — measured Ze	Within the 0.35 Ω typical maximum	0.28 Ω
Measured Zs at the furthest point	So (R ₁ +R ₂) = 1.12 – 0.28	1.12 Ω
Tabulated maximum	Sheet 2, Type B, 32 A	1.37 Ω
Against the tabulated figure	1.12 Ω < 1.37 Ω	Complies
Against Appendix 3 (cold measurement)	0.8 × 1.37 = 1.10 Ω, and 1.12 Ω > 1.10 Ω	Does not

That last pair of rows is the whole point of this sheet. 1.12 Ω sits inside the tabulated 1.37 Ω but outside the 1.10 Ω Appendix 3 accepts for a cold measurement — so a reading that looks compliant against the table does **not** meet BS 7671's criterion. Compare like for like, every time, before you write C1, C2 or a tick.

Maximum Zs — Circuit-breakers & RCDs

BS 7671 Tables 41.3 & 41.5 · BS EN 60898, BS EN 61009, BS 3871 · 230 V · as tabulated, $C_{min} 0.95$ · no temperature correction applied



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Maximum Zs (Ω) — BS EN 60898 / BS EN 61009, 230 V

Rating (A)	3	6	10	16	20	25	32	40	50	63	80	100	125
Type B — $5 \times I_n$	14.57	7.28	4.37	2.73	2.19	1.75	1.37	1.09	0.87	0.69	0.55	0.44	0.35
Type C — $10 \times I_n$	—	3.64	2.19	1.37	1.09	0.87	0.68	0.55	0.44	0.35	0.27	0.22	0.17
Type D — $20 \times I_n$ (0.4 s)	—	1.82	1.09	0.68	0.55	0.44	0.34	0.27	0.22	0.17	0.14	0.11	0.09
Type D — $10 \times I_n$ (5 s)	—	3.64	2.19	1.37	1.09	0.87	0.68	0.55	0.44	0.35	0.27	0.22	0.17

Type B and Type C carry one row each — the figure satisfies both the 0.4 s and the 5 s disconnection time. **Type D is the only curve with two:** it trips at $20 \times I_n$ for 0.4 s but $10 \times I_n$ is enough for 5 s, so the 5 s values are exactly double. That is a genuine shared formula with Type C, not a misprint.

BS 3871-1 circuit-breakers Types 1, 2, 3 and 4

BS 3871-1 is withdrawn, superseded by BS EN 60898 — which is exactly why you need this table on an EICR: these devices are still in service in a great many older boards. **Types 1-4 are not interchangeable with B/C/D.** A Type 1 trips at $4 \times I_n$ where a Type B trips at $5 \times I_n$, so recording an old Type 1 as a "Type B" understates its permitted Zs by 20% and can fail a circuit that complies. BS 7671 no longer tabulates these; the figures below follow the IET On-Site Guide.

Maximum Zs (Ω) — circuit-breakers to BS 3871-1, 230 V

Type	Trips at	5	6	10	15	16	20	25	30	32	40	45	50	60	63	100
Type 1	$4 \times I_n$	10.93	9.1	5.46	3.64	3.41	2.73	2.19	1.82	1.71	1.37	1.21	1.09	0.91	0.87	0.55
Type 2	$7 \times I_n$	6.24	5.2	3.12	2.08	1.95	1.56	1.25	1.04	0.98	0.78	0.69	0.62	0.52	0.5	0.31
Type 3	$10 \times I_n$	4.37	3.64	2.19	1.46	1.37	1.09	0.87	0.73	0.68	0.55	0.49	0.44	0.36	0.35	0.22
Type 4	$50 \times I_n$	0.87	0.73	0.44	0.29	0.27	0.22	0.17	0.15	0.14	0.11	0.1	0.09	0.07	0.07	0.04

RCDs BS 7671 Table 41.5 · Reg 411.5.3

Where disconnection is by residual current rather than overcurrent, the limit is $U_L \div I_{\Delta n}$, with $U_L = 50$ V.

$I_{\Delta n}$	30 mA	100 mA	300 mA	500 mA
Maximum Zs (Ω)	1667	500	167	100

An RCBO is not an RCD for this purpose. It has its own overcurrent element, so its Zs comes from Table 41.3 — a Type B 32 A RCBO is 1.37 Ω, exactly the same as the MCB. The $50/I_{\Delta n}$ limit governs only where a standalone RCD (BS EN 61008) provides the fault protection. ⚠️ Note 2 to Table 41.5 advises that the installation earth electrode resistance should be as low as practicable, and that a value exceeding 200 Ω *may not be stable* — soil dries and freezes. That is advice about stability, not a permitted maximum.

Maximum Zs — Fuses

BS 7671 Tables 41.2 (0.4 s) & 41.4 (5 s) · 230 V · as tabulated, C_{min} 0.95 · no temperature correction applied



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0.4 s disconnection BS 7671 Table 41.2 · maximum Zs in ohms

BS 88-2 / BS 88-6 (gG)

Rating (A)	2	4	6	10	16	20	25	32	40	50	63
Max Zs (Ω)	33.1	15.6	7.8	4.65	2.43	1.68	1.29	0.99	0.75	0.57	0.44

BS 88-3 (system C)

Rating (A)	5	16	20	32	45	63
Max Zs (Ω)	9.93	2.3	1.93	0.91	0.57	0.36

BS 3036 (rewirable)

Rating (A)	5	15	20	30	45	60
Max Zs (Ω)	9.1	2.43	1.68	1.04	0.56	0.4

BS 1362 (plug-top)

Rating (A)	3	13
Max Zs (Ω)	15.6	2.3

5 s disconnection BS 7671 Table 41.4 · maximum Zs in ohms

BS 88-2 / BS 88-6 (gG)

Rating (A)	2	4	6	10	16	20	25	32	40	50	63	80	100	125	160	200
Max Zs (Ω)	44	21	12	6.8	4	2.8	2.2	1.7	1.3	0.99	0.78	0.55	0.42	0.32	0.27	0.18

BS 88-3 (system C)

Rating (A)	5	16	20	32	45	63	80	100
Max Zs (Ω)	14.6	3.9	3.2	1.6	1	0.68	0.51	0.38

BS 3036 (rewirable)

Rating (A)	5	15	20	30	45	60	100
Max Zs (Ω)	16.8	5.08	3.64	2.51	1.51	1.07	0.51

BS 1362 (plug-top)

Rating (A)	3	13
Max Zs (Ω)	22	3.64